CIRRICULUM VITAE

Timothy Belin

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EDUCATION

Union College | Schenectady, NY *Expected Spring 2026*

Major: Mechanical Engineering; Minor: Mathematics

- GPA: 3.98/4.0, Dean's List
- Relevant Coursework: Design of Thermofluidic Systems, Fluid Mechanics, Heat Transfer, Dynamics of Physical Systems, Programming and Writing in Mechanical Engineering, Kinematic Machine Design, Mechanics of Materials 1 & 2, Thermodynamics 1 & 2, Dynamics, Engineering Reliability, Differential Equations, Partial Differential Equations, Applied Linear Algebra, Introduction to Computer Science, Introduction to Circuits.

ENGINEERING RESEARCH

Union College – Professor Ali Hamed | *Research Assistant* | Schenectady, NY *Jun. 2024 – Present*

- Experimentally investigates turbulent boundary-layer flow over various rough surfaces to determine the effects of the roughness geometry on the flow characteristics and structure.
- Employs planar particle image velocimetry (PIV) and volumetric particle tracking velocimetry (PTV) in a recirculating water tunnel and an open wind tunnel for flow field measurements.
- Develops MATLAB code to analyze and visualize the experimental results, including mean flow fields, turbulence quantities, and second-order turbulence statistics.
- Authors journal publication written drafts and assists Professor Hamed on responding to peer-review feedback.

Publications:

[3] A. M. Hamed, W. H. Good, J. A. Carreiro, T. R. Belin, and M. Kaid, "Wake organization and flow relaxation past rigid and flexible isolated roughness elements," **In Preparation**.

[2] T. R. Belin, B. Mlavsky, and A. M. Hamed, "Turbulent flow past a finite wall-mounted cylinder at various degrees of upstream sheltering: Effects of the number of upstream cylinders," *Physics of Fluids*, 37 (11), Art. No. 115158, 2025.

[1] A. M. Hamed, I. A. Spoor, B. R. McAtee, and T. R. Belin, "Vortical structure organization past isolated and sheltered low-aspect-ratio finite wall-mounted circular cylinders immersed in a turbulent boundary layer," *Experiments in Fluids*, 66 (212), Art. No. 212, 2025.

Presentations:

- T. R. Belin, B. Mlavsky, and A. M. Hamed, "Impact of the number of upstream sheltering cylinders on the wake of a downstream finite wall-mounted cylinder," American Physical Society – Division of Fluid Dynamics, Houston, TX (November, 2025).
- B. Mlavsky, T. R. Belin, and A. M. Hamed, "Turbulent flow past a low-aspect-ratio finite wall-mounted cylinder at various degrees of upstream sheltering," American Physical Society – Division of Fluid Dynamics, Salt Lake City, UT (November, 2024).
- T. R. Belin, "Turbulent flow past a low-aspect-ratio finite wall-mounted cylinder at various degrees of upstream sheltering", Steinmetz Symposium Mechanical Engineering Session, Union College, Schenectady, NY (May, 2025).

TEACHING EXPERIENCE

Union College – Mechanical Engineering Dept. | *Supplemental Instructor* | Schenectady, NY *Apr. 2024 – Jun. 2025*

- Supported student learning in Programming and Writing in Mechanical Engineering, a second-year mechanical engineering course teaching library research, scientific writing, and data analysis skills.
- Led review sessions covering MATLAB coding, including visualization of large datasets, numerical differentiation/integration, and numerical solutions to simple differential equations.
- Gave feedback to students in individual appointments on the clarity, conciseness, and adherence to scientific writing conventions within their writing assignments.

Union College – Writing Center | *Peer Tutor* | Schenectady, NY *Sep. 2023 – Mar. 2024*

- Provided guidance on essays, lab reports, and all other forms of written communication.
- Assisted students with organization, argument structure, and formatting.

ENGINEERING PROJECTS

Localized height heterogeneity effects on flow over canopy | *Mechanical Engineering Senior Thesis (in progress)*

- Experimentally investigated the effects of localized canopy height heterogeneity protrusions on the turbulent flow over an overwise homogeneous submerged canopy using a recirculating water channel.
- Lead experimental preparation, including laser alignment, camera alignment, and camera calibration.
- Collected flow measurements using 3D volumetric particle tracking velocimetry (PTV) to capture the three-dimensionality of the coherent vortical structures within the heterogeneity wake.
- Completed mean flow, turbulence statistic, and vortex shedding analyses on the results.

Piping System Design for Fuel Oil Delivery | *Design of Thermofluidic Systems*

- Led a four-person team in designing a cost-effective, code-compliant piping system for delivering No. 5 heavy fuel oil from storage to an industrial furnace.
- Performed thermofluidic, heat transfer, and economic analyses encompassing a range of pipe, insulation, and pump configurations to optimize for performance and cost while adhering to spill prevention standards.

Dual-Degree-of-Freedom Stair-Climbing Wheel | *Kinematic Machine Design*

- Designed and manufactured a wheel capable of climbing steps using two input axles: one for conventional wheel rotation and the other for radial lobe tread extension to provide traction on the elevated step surface.
- Fully modeled and validated the gear and linkage systems in SolidWorks to ensure proper prefabrication component meshing.

LEADERSHIP & ENGAGEMENT

Pi Tau Sigma Mechanical Engineering Honor Society | *Vice President* | Schenectady, NY *2025 - Present*

- Organizes the Union College mechanical engineering help desk, coordinating member volunteer schedules and collaborating with the mechanical engineering department faculty.
- Provides extracurricular academic support to mechanical engineering students on homework, laboratory reports, and other technical assignments.

Union College Ultimate Frisbee Club Team | *President* | Schenectady, NY *Jun. 2025 - Present*

- Coordinates team practices, tournament travel, registration, and logistics.
- Serves as liaison between the team, the Student Activities department, and Finance Committee.
- Fosters a positive team culture by supporting player development and ensuring an inclusive environment on and off the field for all team members.

Union College Outing Club | *President* | Schenectady, NY *Apr. 2025 – Apr. 2026*

- Leads hiking, rock climbing, white water rafting, and mountain biking trips to local outdoor settings.
- Organizes transportation, funding, and community outreach for each outdoor trip in collaboration with the Student Activities Department and the other Outing Club student leaders.

Union College Science and Technology Entry Program | *Volunteer Tutor* | Schenectady, NY *Oct. 2022 - Jun. 2023*

- Tutored math and science for underserved high school students from the local Schenectady school system.

Lafayette Public Library Homework Center | *Volunteer Tutor* | Lafayette, CO *Sep. 2019 - Mar. 2020*

- Tutored local elementary and middle school students in math, science, reading, and writing.
- Provided free academic support for families unable to afford private tutoring.

AWARDS & HONORS

Tau Beta Pi Engineering Honor Society | New York Mu Chapter, Union College *Nov. 2025*

John S. Hadala (1928) Endowed Book Prize | *Awarded* | Union College *May 2025*

- Awarded to the rising senior majoring in mechanical engineering on the basis of academic, personal, and social achievement.

Pi Tau Sigma Mechanical Engineering Honor Society | Alpha Alpha Chapter, Union College *Mar. 2025*

General Electric Edison Mechanical Engineering Book Prize | *Nominated* | Union College *May 2024*

- Awarded to a sophomore majoring in mechanical engineering who has demonstrated exceptional leadership qualities.

OTHER PROFESSIONAL EXPERIENCE

Carve Construction | *General Contractor* | Boulder, CO *Oct. 2021 - Aug. 2022*

- Led and assisted residential remodels with a focus on cabinet refinishing, trim, and finishing work.
- Attentively followed design specifications and verbal instructions to achieve high-quality results.
- Experienced in drywall, tile, electrical, framing, and painting work.

Keewaydin Dunmore Camp for Boys | *Counselor* | Salisbury, VT *Jun. 2023 - Aug. 2023*

- Summer camp counselor mentoring teenage boys in a rustic outdoor setting.
- Led multi-week whitewater canoeing expeditions in Maine and Quebec.
- Fostered the development of maturity, leadership, and healthy masculinity qualities in the campers.
- Taught outdoors skills including navigation, canoeing, sailing, and campcraft.

SKILLS

Software: MATLAB, SolidWorks (Certified Professional), Autodesk Inventor (Certified Associate), Python

Manufacturing: Mills, Lathes, Handheld Power Tools, Nail Guns, Table Saws, Concrete Cutters

EXTRACURRICULARS

Interests: Biking, Running, Ultimate Frisbee, Hiking, Skiing, Camping, Solving Rubix's Cubes